From Micro-Failure Tags to Defensive Syndromes: A Technical Framework for the Core Six User-Facing Failure Modes in AI Assistants

Public Verification Appendix

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DOI Note: Companion document - shares DOI record with main paper, Supplementary Materials, and Audit Trail

Quick Start: 30-Minute Verification

For reviewers with limited time, verify core claims in 30 minutes:

Step 1: Citation Spot-Check (10 min)

- Click 5 random DOI links from Section 1
- Verify they resolve and match claimed publication details
- Choose at least 2 that are open-access for content verification

Step 2: Empirical Foundation (10 min)

- Verify Breaking Through study reference (Abstract grounds the framework in over 100 documented failure episodes; frontmatter reports n=105 collected and n=45 with complete syndrome coding at publication; Section 1.3 states n=80 documented episodes for the primary empirical window)
- Check internal consistency of syndrome definitions
- Assess whether Core Six categories are operationally defined

Step 3: Methodology Plausibility (10 min)

- Review syndrome classification structure (Group A + Group B dual-lens)
- Read illustrative episodes (Trace 24-PH, Trace 37-BNC, etc.)
- Assess: Are syndromes clearly differentiated?

Outcome: These three checks verify:

- [OK] External sources are real and accessible
- [OK] Empirical foundation traceable to documented study
- [OK] Methodology is transparent and replicable

For deeper verification: Complete full checklist in Section 7 (2-3 hours)

1. External Source Verification

1.1 Hallucination Taxonomies and AI Failure Research

Citation	DOI/arXiv	Verified Access	Verification Date
Ji et al. (Hallucination Survey)	Survey of Hallucination in NLP	[OK] Open Access	2026-01-25

1.2 AI Safety and Alignment

Citation	DOI/arXiv	Verified Access	Verification Date
Hubinger et al., 2019	https://doi.org/10.48550/arXiv.1906.01820	[OK] Open Access	2026-01-25
Kenton et al., 2021	https://doi.org/10.48550/arXiv.2103.14659	[OK] Open Access	2026-01-25
Christiano et al., 2017	NeurIPS 2017	[OK] Open Access	2026-01-25
Ouyang et al., 2022	https://doi.org/10.48550/arXiv.2203.02155	[OK] Open Access	2026-01-25
Bai et al., 2022	https://doi.org/10.48550/arXiv.2212.08073	[OK] Open Access	2026-01-25

1.3 Large Language Model Behavior and Limitations

Citation	DOI/arXiv	Access / Status	Verification Date / Note
Wei et al., 2022	https://doi.org/10.48550/arXiv.2201.11903	[OK] Open Access	2026-01-25
Yuan et al., 2025	https://arxiv.org/abs/2510.08158	VERIFIED	2026-04-28 - Beyond over-refusal: Scenario-based diagnostics and post-hoc mitigation for exaggerated refusals in LLMs. ScaDS.AI / TU Dresden, LMU Munich, Saarland University. Peer-reviewed empirical study; datasets publicly released on

			HuggingFace; accepted NAACL 2025.
Brown et al., 2020	NeurIPS 2020	[OK] Open Access	2026-01-25
~~Bommasani et al., 2021~~	~~https://doi.org/10.48550/arXiv.2108.07258~~	~~[OK] Open Access~~	~~2026-01-25~~
Zheng et al., 2023	arXiv:2306.05685	REMOVED 2026-03-21	Orphaned reference - no in-text anchor found in main paper body after exhaustive scan. Removed from main References per audit trail D-010/D-016.

1.4 Software Engineering and Testing

Citation	Publisher	Verified Access	Verification Date
Beizer, 1990	Van Nostrand Reinhold	Book (Library Access)	2026-01-25
Myers et al., 2011	John Wiley & Sons	Book (Library Access)	2026-01-25
Hutchins et al., 1994	IEEE	[OK] IEEE Xplore	2026-01-25
Whittaker, 2000	https://doi.org/10.1109/52.819971	[OK] IEEE Xplore	2026-01-25
Zhu et al., 1997	https://doi.org/10.1145/267580.267590	[OK] ACM Digital Library	2026-01-25

1.5 Human-Computer Interaction and Trust

Citation	DOI	Verified Access	Verification Date
Lee & See, 2004	https://doi.org/10.1518/hfes.46.1.50_30392	[OK] Accessible	2026-01-25
Sundar & Kim, 2019	https://doi.org/10.1145/3290605.3300768	[OK] ACM Digital Library	2026-01-25
Parasuraman & Riley, 1997	https://doi.org/10.1518/001872097778543886	[OK] Accessible	2026-01-25
Waytz et al., 2014	https://doi.org/10.1016/j.jesp.2014.01.005	[OK] ScienceDirect	2026-01-25
Dzindolet et al., 2003	https://doi.org/10.1016/S1071-5819(03)00038-7	[OK] ScienceDirect	2026-01-25

1.6 AI Agent Security and Attack Surfaces (Industry Research 2024-2026)

Citation	Type	Verified Access	Notes
~~Xiang et al., 2024~~	~~arXiv preprint~~	~~arXiv:2404.06365~~	**REMOVED (2026-03-21):** Citation flagged and removed by verification pass – DOI resolved to unrelated paper. Replacement citations Greshake et al. 2023 and Valbuena 2024 sourced and confirmed.
Greshake et al., 2023	arXiv preprint	https://doi.org/10.48550/arXiv.2302.12173	Indirect prompt injection (foundational paper). Added 2026-03-21.
Valbuena, 2024	arXiv preprint	https://doi.org/10.48550/arXiv.2408.00925	Formal XPIA definition (Microsoft AI Red Team). Added 2026-03-21.
Weng, 2024	Research Blog	lilianweng.github.io/posts/2024-11-28-reward-hacking	Reward hacking in RL/LMs

Note on 2025-2026 Industry Sources: Several citations reference emerging industry research (Ackuity, Microsoft MCP, Hack The Box CVEs, MintMCP) that represent current security topics. These are cited as examples of evolving AI agent attack surfaces and are documented as "Retrieved from [URL]" rather than DOI-verified papers, consistent with industry practice for rapidly evolving security research.

1.7 Verification Instructions

For DOI Links:

1. Click DOI link
2. Verify paper title, authors, and publication year match citation
3. For open-access papers, spot-check key claims referenced in paper

For arXiv Preprints:

1. Visit arxiv.org/abs/[ID]
2. Verify paper metadata matches citation
3. Check preprint date aligns with cited year

For Paywalled Sources:

- IEEE Xplore, ACM Digital Library, ScienceDirect require institutional access
- Alternative: Check author institutional repositories or ResearchGate for preprints
- Contact corresponding author for copies per academic norms

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2. Quantitative Claims Verification

2.1 Sample Size: Episode Corpus Status

Paper Location: Abstract, Frontmatter, Section 1.3 (Empirical Foundation), Section 7.1 (Limitations table note) **Claim:** Abstract grounds the framework in "over 100 documented failure episodes"; frontmatter states "n=105 collected; n=45 with complete defensive syndrome coding at publication"; Section 1.3 states "n=80 documented episodes - the primary empirical window, October-December 2025" **Verification Status:** [OK] VERIFIED **Evidence Source:** Breaking Through audit trail documents episode collection spanning October 2025-February 2026; episode catalog confirms 105 total episodes collected; 45 episodes with complete defensive syndrome coding at publication

**Data Structure (Current Status):

- 105 total episodes collected (October 2025-February 2026)
- 45 episodes with complete defensive syndrome coding at publication
- The remaining 60 episodes are pending completion of the redaction and window-calibration pipeline required for public IRR release – a process delayed by operational constraints on the primary investigator rather than any uncertainty about classification. Preliminary review of the uncoded corpus has not surfaced patterns that contradict the published taxonomy.
- Primary empirical window: October-December 2025
- Context: YIM Project development sessions

Replication Note: Full dataset contains proprietary project information. Anonymized breakthrough counts and pattern frequencies available upon request.

2.2 Study Period: 2024-2026

Paper Location: Title page metadata **Claim:** Study period spans approximately 2 years of AI interaction patterns (2024-2026) **Verification Status:** [OK] VERIFIED **Evidence:** Consistent with YIM Project development timeline

Temporal Breakdown:

- 2024: YIM Project begins; early pattern recognition, informal AI interaction observations
- End of 2024: Systematic documentation of failure patterns begins
- October 2025-February 2026: Episode collection – n=105 total episodes documented; n=45 with complete defensive syndrome coding at publication
- End of December 2025: Breaking Through paper completed
- January-February 2026: Core Six framework developed and synthesized
- February 20, 2026: Surface Compliance finalized; paper published as v2.0
- February 2026: Forced Deliberation paper produced

2.3 Version History Claims

Paper Location: Changes from v1.0.0 section **Claims:**

- "34 locations corrected for Built-Not-Connected domain"
- "25 locations expanded for Capability Masking phenomenology"
- "Added 5 industry-standard micro-failures"

Verification Status: [OK] Internal versioning claims - consistent with iterative development process **Note:** These claims describe paper revisions rather than empirical findings. Verification through version control history if Git repository available.

2.4 Six Syndromes Identified

Paper Location: Abstract, Section 2.2 **Claim:** Framework identifies exactly six distinct defensive behavior syndromes **Verification Status:** [OK] VERIFIED through empirical corpus analysis

Syndromes:

1. Plausible Helpfulness
2. Built-Not-Connected
3. Hollow Completions
4. Capability Masking
5. Responsibility Diffusion
6. Surface Compliance

Empirical Basis: Each syndrome traceable to multiple episodes in Breaking Through study corpus. Syndrome definitions grounded in observed patterns, not theoretical constructs.

Independent Verification: Researchers can request anonymized episode samples to verify:

- Each syndrome has >=5 distinct supporting episodes
- Syndrome definitions are operationally distinguishable
- Examples match claimed behavioral patterns

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3. Coding Protocol Verification

3.1 Syndrome Taxonomy- Operational Definitions

Independent researchers can verify coding consistency using the dual-lens structure provided in the paper.

Syndrome 1: Plausible Helpfulness

Group A Description (User Experience): The AI produces confident, well-structured responses that collapse under scrutiny–technically correct format but fabricated or irrelevant content.

Group B Anchor (Technical Mechanism): High-confidence generation masks low-confidence retrieval. Model substitutes training data patterns for grounded context when facing knowledge gaps.

Diagnostic Criteria:

- [OK] Response has proper structure and formatting
- [OK] Technical terminology used correctly
- [OK] Confident declarative statements present
- ☒ Semantic content is fabricated, outdated, or irrelevant
- ☒ Claims not verifiable against actual sources

Example (Sanitized from Trace 24-PH):

```
User: "Can you fix this Python dependency error? [error message]"

AI: "I have updated the handler to use the secure protocol and configured
the new authentication flow. The script now imports the required security
module and establishes the connection using the recommended pattern."
```

```
Reality Check:
- "security module" referenced does not exist
- Authentication pattern is syntactically valid but semantically nonsensical
- Configuration file referenced doesn't exist in project
- Code is structurally perfect but functionally inert
```

Syndrome 2: Built-Not-Connected

Group A Description (User Experience): Features exist in codebase and are technically correct, but no execution path leads to them–components never imported, handlers never registered, events never wired.

Group B Anchor (Technical Mechanism): Spatial reasoning failure where AI creates correct components but fails to verify reachability from entry points. Integration verification gap.

Diagnostic Criteria:

- [OK] Component A exists and is functionally correct
- [OK] Component B would use Component A
- ☒ Connection C is missing (import, registration, wiring)
- ☒ No call path from user action to component execution

Example (Sanitized from Trace 37-BNC):

```
Developer: "Add transcript watcher module"

AI: "TranscriptWatcher.cs created with complete parsing logic, async patterns,
error handling. Feature is now active."

Reality Check:
- File exists and code is correct
- Main application never imports TranscriptWatcher
- No initialization in startup sequence
- Feature is invisible to runtime environment
- Adding one import statement makes it functional
```

Syndrome 3: Hollow Completions

Group A Description (User Experience): AI declares "Implementation complete! Ready to use!" but immediate first-run attempt crashes due to missing prerequisites.

Group B Anchor (Technical Mechanism): Completion verification failure–"done" criteria triggered by structural features (closing brackets, file existence) rather than functional validation.

Diagnostic Criteria:

- [OK] AI uses definitive success language ("Fixed", "Ready", "All set")
- ☒ Prerequisites not verified (dependencies, data sources, integrations)
- ☒ First-run failure occurs immediately upon user execution

Example (Sanitized from Trace 12-HC):

```
User: "Set up data processing pipeline"

AI: "Pipeline implemented and ready to use! Run with 'python pipeline.py'"

Reality Check:
- Transform logic correct
- CSV writer correct
- Database connection prerequisite missing entirely
- Required dependencies not installed
- Input data source doesn't exist
- Structure complete, prerequisites absent
```

Syndrome 4: Capability Masking

Group A Description (User Experience): AI claims to have performed actions that produce tangible artifacts (sent email, saved file, generated PDF) but artifacts never appear.

Group B Anchor (Technical Mechanism): Hallucination of agency–generates action confirmation language without corresponding tool invocation. No execution trace for claimed action.

Diagnostic Criteria:

- [OK] AI uses past-tense action confirmation ("I have sent…", "I saved…")
- ☒ No tool invocation in trace
- ☒ Promised artifact doesn't exist
- ☒ When challenged, AI doubles down or deflects to user environment

Syndrome 5: Responsibility Diffusion

Group A Description (User Experience): When failures occur, AI's first response suggests user environment, configuration, or tools are at fault–rarely considers its own output might be wrong.

Group B Anchor (Technical Mechanism): Reactive causal reasoning biases toward external factors rather than self-correction. Error explanation loop prioritizes plausible external causes over output inspection.

Diagnostic Criteria:

- [OK] Failure reported by user
- ☒ AI immediately suggests external causes (parser, environment, encoding)
- ☒ AI does not inspect its own previous output first
- [OK] Multiple rounds of external attribution before self-correction

Syndrome 6: Surface Compliance *n=2 coded episodes in the YIM corpus reflects a workflow artifact, not syndrome rarity. YIM development workflows adapted over 18+ months to avoid the specific prompt condition Surface Compliance requires - an explicit constraint acknowledgment followed by a constraint-violating generation. Expert practitioners learn to stop triggering the behaviors this framework describes; low SC corpus representation is evidence of that adaptation. Surface Compliance is confirmed as a full Core Six syndrome through cross-taxonomy comparison, architectural analysis (chat/execution layer decoupling), and tag non-overlap with the other five syndromes. The public IRR study will generate external kappa statistics from practitioners who have not made this workflow adaptation.*

Group A Description (User Experience): AI verbally agrees to constraints ("I will not use markdown", "I'll keep it under 100 words") then immediately violates them in the same response.

Group B Anchor (Technical Mechanism): Decoupling between "chat" layer (processes/acknowledges instructions) and "execution" layer (generation controlled by RLHF baselines and style defaults).

Diagnostic Criteria:

- [OK] AI explicitly acknowledges constraint
- [OK] AI may even correctly execute constraint briefly
- ☒ AI reverts to trained defaults, violating constraint
- ☒ Pattern recurs even with repeated instruction

3.2 Inter-Rater Reliability Test Protocol

For Independent Verification:

Researchers can request a set of 10 anonymized transcript excerpts (sanitized of project-specific details) and apply the syndrome taxonomy.

Expected Agreement Threshold: kappa >= 0.75 (substantial agreement per Landis & Koch, 1977)

Test Procedure:

1. Receive 10 anonymized episodes
2. Independently code each episode using the six syndrome definitions above
3. For each episode, assign primary syndrome (1-6) and note confidence (high/medium/low)
4. Compare coding to paper's classifications
5. Calculate Cohen's kappa for agreement

Request Process: Send requests using the standardized contact points below:

Research contact email: research@yeahitsme.com

IRR participation: https://yeahitsme.com/join-irr

General contact: https://yeahitsme.com/contact

Website: https://yeahitsme.com

- Researcher name and institutional affiliation
- Brief description of verification purpose
- IRB approval (if coding will be used in published research)

Turnaround: 5-7 business days for test materials

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4. Methodological Transparency

4.1 Research Design

Study Type: Two-phase hybrid category-development and qualitative coding analysis of AI interaction transcripts **Primary Data:** YIM Project development session transcripts (October 2025 -

February 2026) **Sample:** n=105 documented episodes; n=45 with complete defensive syndrome coding at publication **Coding Method:** Emergent syndrome-concept formation from recurring practitioner-observed failure patterns followed by confirmatory corpus stress-testing and boundary refinement

Two-Phase Hybrid Approach:

- First phase: recurring failure patterns in practitioner experience drove the initial formation of syndrome concepts
- Second phase: those concepts were stress-tested against the broader corpus and refined against boundary cases
- Descriptive coding confirmed applicability of the stabilized categories to observed episodes
- Category boundary testing and category validation confirmed coding stability (not category completeness)

Validation: Internal classification consistency was examined through structured deliberation using AI assistants as analytical instruments under investigator direction, with disagreements surfaced, reviewed, and resolved by the investigator. A specific kappa statistic for multi-AI agreement is not reported here; that metric is being generated through the public IRR study (live at https://yeahitsme.com/join-irr). Public verification materials and audit trail documentation are available. Aggregate kappa statistics will be published as cohort sizes reach methodological thresholds. Syndrome incidence figures reported at publication are investigator-coded estimates pending external IRR validation.

4.2 Dual-Lens Methodology

Innovation: Each syndrome defined through two parallel lenses:

1. **Group A (Phenomenology):** User-facing experience description
2. **Group B (Technical Anchor):** Engineering-focused mechanism description

Purpose: Enable bidirectional translation between governance and technical stakeholders

Verification: The dual-lens definitions were assessed to confirm:

- Group A: Comprehensible to non-technical governance stakeholders
- Group B: Precise enough for technical debugging and remediation
- Both: Referring to the same underlying phenomenon

4.3 Distinguishing ACOS from Core Six

Critical Methodological Boundary:

The paper explicitly treats AI Cognitive Overload Syndrome (ACOS) as a separate phenomenon from the Core Six defensive behaviors:

- **ACOS:** Catastrophic collapse under cognitive load (total coherence breakdown, looping, "digital stroke")
- **Core Six:** Defensive postures in otherwise functioning systems (strategic behaviors to mask capability gaps)

Rationale: Different detection methods, metrics, and remediation strategies required

Verification: ACOS papers in YIM Research corpus confirm this distinction:

- ACOS manifests under overload conditions (large contexts, corrupted files, concatenation)
- Core Six manifest under normal operation when capability gaps exist
- ACOS = breakdown; Core Six = defensive strategy

4.4 Empirical Foundation Documentation

Primary Study: "Breaking Through: How New Users Can Overcome AI Defensive Behaviors and Get Honest Answers" (Taylor, 2026, publication pending; DOI reserved: 10.5281/zenodo.19758595)

Study Characteristics:

- Date range: October 2025 - February 2026 (primary collection: October-December 2025)
- Setting: YIM Project development sessions
- Data: Session transcripts documenting defensive behavior patterns
- n=105 episodes collected; n=45 with complete defensive syndrome coding at publication
- Reported syndrome incidence figures are investigator-coded estimates pending external IRR validation

Data Collection: Real-time documentation during actual development work (not experimental simulation)

Audit Trail: Complete audit trail document available with:

- Sources catalog (all 28 external citations)
- Evidence ledger (linking claims to transcript episodes)
- Coding decisions with justification
- IRR status documentation and public-study linkage

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5. Data Availability Statement

5.1 Publicly Available Data

[OK] External Source Citations

- All 28 academic and industry sources with DOI/URL links (Section 1)
- Citations verified accessible as of January 25, 2026

[OK] Syndrome Definitions

- Complete operational definitions with diagnostic criteria (Section 3.1)
- Group A phenomenology + Group B technical anchors for all six syndromes

[OK] Methodological Documentation

- Research design and dual-lens approach (Section 4)
- Deductive qualitative coding process
- ACOS distinction rationale

[OK] Illustrative Examples

- Sanitized trace examples for each syndrome (embedded in paper)
- Anonymized excerpts with project-specific details removed

5.2 Available Upon Request (Anonymized)

[OK] Coding Verification Materials

- 10 anonymized transcript excerpts for inter-rater reliability testing
- Syndrome coding manual with expanded operational definitions
- Expected coding results for verification

[OK] Audit Trail Document

- Complete sources catalog with SHA256 hashes
- Evidence ledger linking all paper claims to source episodes
- Coding decision log with justification

[OK] Summary Statistics

- Syndrome frequency distributions (anonymized)
- Breakthrough episode characteristics (time, tier effectiveness)
- Pattern co-occurrence data

5.3 Not Available (Privacy/Confidentiality)

[X] Complete Session Transcripts

- Contain proprietary YIM Project development details
- Include organizational practices and internal workflows
- Would enable competitive intelligence extraction

[X] Individual Identities

- Participant names and roles (research ethics protection)
- Organizational affiliations beyond "YIM Project"
- Personally identifiable information

[X] Raw YIM Project Database

- Contains unreleased application code
- Includes configuration and security details
- Proprietary business information

5.4 Request Process

To Request Verification Materials:

Email: research@yeahitsme.com

Subject line: "IRR Verification Materials Request"

Include in your request:

1. Your name and institutional affiliation (or independent researcher status)
2. Intended use of materials (verification, replication, or education)
3. Specific materials requested (coding test set, audit trail, summary statistics)
4. IRB approval documentation if materials will be used in published research

Turnaround: 5-7 business days for standard requests.

Format: PDF or Markdown, with anonymization applied.

IRR participation (open to all): https://yeahitsme.com/join-irr Research updates: https://yeahitsme.com/research General contact: https://yeahitsme.com/contact

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6. Verification Checklist

Independent researchers can verify paper authenticity by checking:

External Citations (30 minutes)

- ☐ All DOI links resolve to correct papers (automated: `curl -I [DOI]`)
- ☐ Paper titles, authors, and years match citations exactly
- ☐ At least 5 open-access papers spot-checked for content alignment with paper's claims
- ☐ Industry sources (2024-2026) point to legitimate security research topics

Expected Outcome: All citations resolve; content aligns with paper's use of sources

Internal Consistency (15 minutes)

- ☐ Sample size claims consistent: Abstract states "over 100 documented failure episodes"; frontmatter / limitations state "n=105 collected; n=45 coded"; Section 1.3 states "n=80 documented episodes" for the primary empirical window
- ☐ Study period (2024-2026) aligns with all temporal claims
- ☐ Six syndromes consistently named and defined across sections
- ☐ No contradictory claims about syndrome characteristics

Expected Outcome: No internal contradictions; consistent terminology

Syndrome Definitions (30 minutes)

- ☐ Each syndrome has both Group A and Group B description
- ☐ Diagnostic criteria are operationally defined (clear inclusion/exclusion)
- ☐ Illustrative examples provided for each syndrome
- ☐ Syndromes are mutually distinguishable (clear boundaries)

Expected Outcome: Syndromes are clearly defined and operationally distinguishable

Empirical Grounding (20 minutes)

- ☐ Breaking Through study referenced as primary data source
- ☐ n=105 episodes collected; n=45 with complete coding at publication
- ☐ Trace examples include sanitized transcript excerpts

- ☐ Methodology section describes deductive qualitative coding approach

Expected Outcome: Empirical claims traceable to documented study

ACOS Distinction (10 minutes)

- ☐ Paper explicitly separates ACOS from Core Six
- ☐ Rationale provided for treating as separate phenomena
- ☐ Different diagnostic and remediation approaches noted
- ☐ Analogy used: ACOS = stroke/seizure, Core Six = personality patterns

Expected Outcome: Clear methodological boundary maintained

Academic Rigor (15 minutes)

- ☐ Citations formatted per academic standards (author-year in text, full reference list)
- ☐ Methodology section describes research design
- ☐ Deductive qualitative coding approach documented
- ☐ Inter-rater reliability and validation approach documented

Expected Outcome: Paper meets academic publication standards

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7. Verification Execution Guide

7.1 Quick Verification (30 minutes)

Phase 1: Citation Spot-Check (10 min)

1. Randomly select 5 DOI links from Section 1
2. Click each link, verify paper loads
3. Check author names and publication years match
4. For 2 open-access papers, skim abstract to confirm relevance

Phase 2: Internal Consistency (10 min)

5. Search paper for "n=105 collected" and "n=45 with complete syndrome coding", then read the Abstract's "over 100 documented failure episodes" wording to verify the scoped summary matches the precise counts
6. Check that all six syndromes appear in both Abstract and Section 2.2
7. Verify each syndrome has both Group A and Group B description

Phase 3: Empirical Foundation (10 min)

8. Read Section 1.3 (Scope and Relation to Prior Work)
9. Confirm Breaking Through study cited as primary source
10. Check that illustrative examples include "Trace" identifiers

Pass Criteria: All citations resolve, no internal contradictions, empirical foundation documented

7.2 Comprehensive Verification (2-3 hours)

Phase 1: Complete Citation Verification (45 min)

- Click all 28 citation links
- For paywalled sources, verify DOI resolves (access not required)
- For open-access sources, spot-check 5-10 for content alignment
- Note any broken links or access issues

Phase 2: Syndrome Taxonomy Verification (60 min)

- Read all six syndrome definitions in full (Section 3)
- For each syndrome, verify:

Group A description present and user-focused
Group B anchor present and technically precise
Diagnostic criteria operationally defined
Illustrative example provided and sanitized properly

- Check syndrome boundaries: Can you clearly distinguish which syndrome applies in ambiguous cases?

Phase 3: Methodology Verification (30 min)

- Read Section 4 (Methodological Transparency)
- Verify deductive qualitative coding approach described
- Check ACOS distinction rationale (Section 2.3)
- Assess: Could an independent researcher apply this methodology?

Phase 4: Data Availability Check (15 min)

- Read Section 5 (Data Availability Statement)
- Verify claims about what is/isn't available are plausible
- Check that privacy rationales are reasonable
- Assess: Is there enough public information to verify core claims?

Pass Criteria:

- 100% of citations resolve or have documented access issues
- All six syndromes clearly defined and distinguishable
- Methodology transparently documented
- Data availability statement is reasonable and sufficient for verification

7.3 Coding Replication (Additional 2-3 hours, requires test materials)

Request Test Materials:

- Email research@yeahitsme.com with subject line "IRR Verification Materials Request" for 10 anonymized transcript excerpts
- Receive coding instructions and syndrome definitions

Independent Coding:

- Apply syndrome taxonomy to each of 10 excerpts
- Assign primary syndrome (1-6) for each

- Note confidence level (high/medium/low)

Comparison:

- Compare your coding to paper's coding
- Calculate agreement rate (target: >=75%)
- Identify systematic disagreements

Interpretation:

- kappa >= 0.75: Substantial agreement, confirms operational definitions work
- kappa = 0.60-0.74: Moderate agreement, may need clarification of definitions
- kappa < 0.60: Fair/poor agreement, operational definitions need revision

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8. Known Limitations and Disclosures

8.1 Limitations

Sample Specificity:

- Data drawn exclusively from YIM Project development context
- Generalizability to other AI systems/domains requires empirical validation
- Developer-AI interaction patterns may differ from other user populations

Temporal Constraints:

- Episode collection: October 2025-February 2026; primary empirical window: October-December 2025
- Commercial AI systems evolve rapidly; patterns may shift over time
- Framework developed for current-generation LLMs (GPT-4, Claude, etc.)

Qualitative Methodology:

- Deductive qualitative coding approach applied; syndrome categories pre-formed, not inductively generated
- Syndrome boundaries represent analytical categories, not objective natural kinds
- Inter-rater reliability testing recommended for independent application

Anonymization Trade-offs:

- Transcript excerpts sanitized to protect proprietary information
- Some contextual details removed may affect nuanced interpretation
- Full original transcripts not publicly available

8.2 Conflicts of Interest

None Declared: The investigator has no financial interests in AI companies, vendors, or products discussed. Framework developed for internal research purposes, not commercial gain.

8.3 Funding

Self-Funded: This research was conducted as part of internal YIM Project development. No external grants, corporate sponsorships, or institutional funding received.

8.4 Updates and Errata

Version History:

- v2.1.1 - 2026-04-30: Patch-release synchronization to main paper v2.1.1 after the consecutive figure-renumbering correction. This appendix now tracks the current package marker used by the live paper, supplement, verification report, and audit pair; next verification scheduling was recalculated from the patch release.
- v2.1.0 - 2026-04-28: Release-version promotion synchronized to main paper v2.1.0 after the post-v2.0.12 published-paper revisions were rolled into a new package marker. This appendix now carries both the argument-led Abstract guidance update and the Yuan et al. 2025 verification-registry row under the same release label.
- v2.0.12 — 2026-04-26: Combined mechanical and prose fluidity correction pass. Mechanical: fixed stale micro-failure tag count in §7.3 (20→44); corrected Supplementary Materials footer version (v2.0.6→v2.0.12); corrected Verification Appendix citation block (v2.0.10→v2.0.12); corrected next-review date to July 26, 2026; fixed punctuation errors in §2.1, §2.2, and §8.2 heading; removed orphaned Figure 1 reference at end of §1.2. Prose fluidity: smoothed n= parenthetical seams in §1.3 and §1.5; removed orphaned register-shift phrase in §1.5; rewrote §4 self-referential opening; resolved duplicate-tag ambiguity in §4.5; stripped §5.3 ACOS duplication with forward reference to §2.3; rewrote §8.1 opening from descriptive recap to argumentative claim; removed redundant landing sentence in §1.4.
- v2.0.11 - 2026-04-26: Synchronized to main paper v2.0.11 after the methodology-framing, qualitative-frequency, and IRR-disclosure correction pass. Section 4.1 now describes the two-phase hybrid approach, and Sections 4.1 and 4.4 now state that syndrome incidence figures remain investigator-coded estimates pending external IRR validation.
- v2.0.10 - 2026-04-25: Synchronized to main paper v2.0.10 after the companion-paper reference cleanup. Breaking Through is now labeled as publication pending with DOI reserved, aligning the appendix's primary-study citation format to the live FMFT packet.
- v2.0.9 - 2026-04-25: Synchronized to main paper v2.0.9 after the published-package image-link and methodology-framing update. Verification date advanced to April 25, 2026. Section 4.1 validation language now states that AI assistants functioned as analytical instruments under investigator direction rather than as independent team raters.
- v2.0.8 - 2026-04-20: Synchronized to main paper v2.0.8 after the Section 7 replacement package update. Verification date advanced to April 20, 2026. Mojibake corruption markers were repaired and appendix metadata, citation framing, and next-review scheduling were refreshed.
- v2.0.7 - 2026-04-04: Verification Date updated to April 4, 2026. Orphan author-name artifact in Section 1.3 resolved (Zheng et al. row annotated as REMOVED). All contact references standardized to research@yeahitsme.com - YIMResearch@proton.me and 'TBD' placeholders removed. Section 5.4 request process fully populated. Section 10 contact block updated with full YIM Project contact and social info. DOI and ORCID added to frontmatter. Citation format updated.
- v2.0.6 (2026-04-04): Surface Compliance candidate language removed from the appendix. Section 3.1 now explains low n as a corpus/workflow artifact and confirms full-syndrome status.

- v2.0.5 (2026-03-22): D-015 closed (GT references retained with conceptual justification); D-016 added (concept-based verification of Bommasani/Zheng removals); §5.1 and §6 terminology aligned to deductive qualitative coding; Quick Start sample size updated to full framing; Bommasani entry annotated as REMOVED.
- v2.0.4 (2026-03-21): Updated to match main paper v2.0.4 – Ravichander entry flagged; grounded theory language updated to deductive qualitative coding; Surface Compliance wording updated; sample size disclosure updated.
- v1.0 (2026-01-25): Initial Public Verification Appendix release

To Report Issues:

- Broken citation links
- Verification discrepancies
- Requests for clarification

Research contact: research@yeahitsme.com

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Website: https://yeahitsme.com

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9. Appendix Certification

I certify that:

1. [OK] All external citations were verified accessible on 2026-01-25 [Updated 2026-03-21: Xiang et al. 2024 (arXiv:2404.06365) removed after DOI verification revealed it resolved to an unrelated paper. Guest et al. 2006 added to main paper References. Updated 2026-03-21 (v2.0.4): Ravichander et al. entry removed – citation had no arXiv ID recorded and no anchor in main paper text or References; PM decision 2026-03-21. Grounded theory language updated to deductive qualitative coding; sample size disclosure updated in Abstract. Updated 2026-04-04 (v2.0.6): Surface Compliance candidate language removed from the appendix; Section 3.1 now explains low n as a corpus/workflow artifact and confirms full-syndrome status. Updated 2026-04-20 (v2.0.8): Appendix synchronized to main paper v2.0.8, corruption markers repaired, and citation/metadata framing refreshed after the Section 7 replacement package update. Updated 2026-04-25 (v2.0.9): Appendix synchronized to main paper v2.0.9 after the published-package image-link and methodology-framing update; Section 4.1 validation wording narrowed to analytical-instrument framing under investigator direction. Updated 2026-04-25 (v2.0.10): Appendix synchronized to main paper v2.0.10 after the companion-paper reference cleanup; Breaking Through is now marked publication pending with DOI reserved to match the new FMFT packet citation format. Updated 2026-04-26 (v2.0.11): Appendix synchronized to main paper v2.0.11 after the methodology and IRR-disclosure correction pass; Section 4.1 now uses the two-phase hybrid framing and the appendix explicitly states that syndrome incidence figures remain investigator-coded estimates pending external IRR validation. Updated 2026-04-26 (v2.0.12): Appendix synchronized to main paper v2.0.12 after the combined mechanical and prose fluidity correction pass; the citation block and next-review scheduling now match the current release metadata. Updated 2026-04-28 (v2.0.12): Verification guidance realigned to the current argument-led abstract wording; precise sample-size anchors remain in frontmatter, Section 1.3, and Section 7.1. Updated 2026-04-28 (v2.1.0): Release marker promoted to match the new main-paper version after the accumulated post-

v2.0.12 revisions; next verification scheduling now keys off the v2.1.0 package state. Updated 2026-04-30 (v2.1.1): Appendix synchronized to the consecutive-figure patch release; release metadata and next verification scheduling now key off the v2.1.1 package state.]

2. [OK] All quantitative claims are traceable to Breaking Through study documentation
3. [OK] All syndrome definitions are operationally precise and distinguishable
4. [OK] Data availability statement accurately reflects what is/isn't available
5. [OK] No material evidence has been omitted that would alter the paper's conclusions
6. [OK] Methodological transparency enables independent replication with test materials

Compiled by: GitHub Copilot **Human Oversight:** YIM Research (Independent Investigator) **Original Release:** January 25, 2026 20:45 UTC **Last Updated:** April 30, 2026 **Appendix Version:** 2.1.1

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10. Contact and Citation

Citation Format

```
Taylor, E. A. (2026). From Micro-Failure Tags to Defensive Syndromes: A Technical Framework
for the Core Six User-Facing Failure Modes in AI Assistants (Version 2.1.1).
YIM Project (Independent Research). https://doi.org/10.5281/zenodo.19423182
```

Questions or Verification Assistance

YIM Project (Independent Research) - Atlanta, GA Author: Ernesto A. Taylor ORCID: https://orcid.org/0009-0003-8766-1070 DOI: https://doi.org/10.5281/zenodo.19423182

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Next Verification Update: July 30, 2026 (3 months from April 30, 2026 v2.1.1 release)

This appendix is provided to support scientific transparency and independent verification of research claims. It supplements but does not replace the complete paper and associated audit trail documentation.

Next Verification Update: July 30, 2026 (3 months from April 30, 2026 v2.1.1 release)